

Variables on Both Sides



Grade 8 • Mathematics
NC.8.EE.7

Solve and classify linear equations.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

Solve $12x + 4 = 7x + (-1)$. Show your work and check.

Subtract $7x$ and 4 : $5x = -5$. Divide by 5 : $x = -1$. Both original sides equal -8 .



Build a second model together

Solve $11x + 5 = 8x + (35)$. Show your work and check.

Discuss: What should the first step find?



Your turn • 1

Solve $7x + 6 = 7x + 6$. Classify the solution set and justify it.

Represent → solve → explain → check



Your turn • 2

Solve $6x + 2 = 6x + 5$. Classify the solution set and justify it.

Represent → solve → explain → check



Your turn • 3

Solve $14x + 8 = 9x + (38)$. Show your work and check.

Represent → solve → explain → check



Your turn • 4

Solve $15x + 3 = 8x + (-18)$. Show your work and check.

Represent → solve → explain → check



Exit question • Explain independently

Solve $5x + 7 = 3x + (17)$. Show your work and check.

Include a check. What would you teach someone else?



Teacher review and next steps

1. All real numbers

Exit: $x = 5$

2. No solution

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

3. $x = 6$

4. $x = -3$

